BAKTIAR WASIR FAROOQ

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EDUCATION

B.Sc (Honors) with Research in Physics April 2023 - April 2027
Kirori Mal College, University of Delhi

RELEVANT PUBLICATIONS AND ARXIV PREPRINTS:

1. B. S. Koranga, **B. W. Farooq** and P. Kumar "Modified entanglement patterns in two-flavor neutrinos from quantum-gravity interactions," *Nucl. Phys. B* **1013**, 117373 (2026),doi.
2. B. S. Koranga and **B. W. Farooq** "Two flavor neutrino oscillation in matter and quantum entanglement," *Int. J. Quantum Inf.* **23**, 2540009 (2025), doi.
3. **B. W. Farooq**, B. S. Koranga and M. Blasone "Probing Quantum Foundations in Long-Baseline Neutrino Oscillations: Wave Packet Approach from MINOS Data to DUNE Predictions," arXiv preprint (2026), doi. (*Under Review in Journal*)
4. **B. W. Farooq**, B. S. Koranga and A. Chatterjee "Quantum Fisher information as a probe of sterile neutrino new physics: Geometric advantage of KM3NeT over IceCube," arXiv preprint (2026), doi. (*Under Review in Journal*)

RESEARCH EXPERIENCE

International Project : 2025 - Present
Università degli Studi di Salerno, Italy
External Supervisor: Prof. Massimo Blasone **Internal Supervisor: Prof. Bipin S. Koranga**

- Worked on a project titled **"Probing Quantum Foundations in Long-Baseline Neutrino Oscillations: Wave Packet Approach from MINOS Data to DUNE Predictions"**.
- Derived the wave-packet oscillation probability from first principles, fit it against MINOS data to resolve a known low-energy discrepancy, propagated it to DUNE with realistic matter effects and mass ordering, and used Fisher information to project where the σ_x is best constrained, deriving the triality relations $(\mathcal{P}^2, \mathcal{V}^2, \mathcal{E}^2)$ for detectors sensitive to $|\nu_\mu\rangle$, full pipeline from derivation to data-fitting to discovery-reach projection.

Undergraduate Research Assistant: 2024 - Present
Kirori Mal College, University of Delhi
Supervisor: Prof. Bipin Singh Koranga

- Derived Planck-scale quantum-gravity corrections to GUT-scale neutrino mass-squared splittings and mixing angles, and traced their imprint on the von Neumann entanglement entropy of two-flavor oscillations. Results published in **Nuclear Physics B** (2026). This gave me experience treating quantum-gravity effects as a controlled perturbation on top of GUT-scale seesaw physics and connecting it to an observable entanglement signature.
- Investigated matter effects on two-flavor neutrino oscillations using the Von Neumann entropy of the reduced flavor density matrix, comparing entanglement evolution in matter versus vacuum. Results published in **International Journal of Quantum Information** (2025). This gave me experience deriving matter-modified oscillation parameters and reading coherence effects directly off entanglement measures.
- Applied Quantum Fisher Information and Classical Fisher Information to sterile-neutrino MSW and NSI scenarios, proving exact QFI–CFI saturation for the NSI parameter and quantifying KM3NeT's geometric quantum-information advantage over IceCube via the Cramér–Rao bound. Preprint: arXiv:2604.01256. This taught me to connect abstract quantum-estimation bounds to concrete detector-geometry and baseline-optimization studies, a skill directly transferable to experimental sensitivity design.

Collaboration Project: September, 2026 - Present
Indian Institute of Science (IISc), Bengaluru, India
Collaborator: Dristii Patel (3rd Year B.S Student)

- We developed Python programs to study the tomography effects while using the PREM Model to understand the earth layer density effects in neutrino oscillations and tried to analyze the Neutrino Asymmetries and Violation studies such as CP, CPT and T. *(We are still working on this project.)*

Mini Project at VSoN10: 14 July,2026 - 24 July,2026
International Center for Interdisciplinary Science and Education (ICISE), Vietnam
Supervisor: Dr. Jake McKean

- Worked on a group project on **Neutrino Energy Reconstruction Using Lepton Kinematics vs. Hadron Information**; used **NEUT** to simulate events and generate energy datasets. Simulated CCQE and non-CCQE events, reconstructing neutrino energy via the lepton-kinematic method and comparing E_ν^{rec} against E_ν^{true} to quantify FSI and binding-energy effects.

Research Intern (Semester 5 and 6): 2025 - 2026
Jawaharlal Nehru University, India
External Supervisor: Prof. Ng. K. Francis **Internal Supervisor: Prof. Bipin S. Koranga**

Semester projects (Internships as a part of Skill Enhancement Couse) on **quantum information and CP violation in vacuum neutrino oscillations** (Sem 5) and the **wave-packet approach to oscillations** (Sem 6). Built a Python model of three-flavor vacuum oscillation probabilities, examining δ_{CP}, the Jarlskog invariant, and entanglement entropy as a measure of mixedness in $|\nu_\alpha(t)\rangle$, and analyzed wave-packet decoherence length to understand where the plane-wave approximation fails and how that connects to assumptions in experiments such as MINOS and DUNE.

WORKSHOPS & INVITED TALKS

- **Invited Speaker** on **Quantum Correlations in Neutrino Oscillations** by Dr. Sara BOLOGNESI
  CERN, Zurich (online) **10–11 Dec 2026**
  Talk: *"Wave Packet Approach from MINOS Data to DUNE Predictions"* (25 min + 5 min Q&A).
- **Participant**, **PHLEX: Next-Generation Framework for DUNE and AI/ML Libraries in HEP**,
  IISc Bengaluru **26–30 Oct 2026**
  Five-day workshop with technical discussions and hands-on sessions on DUNE software frameworks and AI/ML libraries in HEP.

HONORS AND ACHIEVEMENTS

- **UNISA Scholarship Recipient with an amount of 8000 euros (Sole International Selection)**, accepted for M.Sc. in Physics at Università degli Studi di Salerno. (Not Availed)
- Awarded **two times** the **Letter of Appreciation by Principal of Kirori Mal College** for academic excellence in research publications - 2026.
- **Selected member of KMC Research Recognistion Certificate Students (KRRCS)-2025**. *(This award is given for Publication in Scopus and SCIE Indexed Journals)*
- **10th Vietnam School on Neutrinos (VSoN) 2026** — Selected as one of 25 international participants (by application) for a two-week program on experimental neutrino physics at IFIRSE, Quy Nhon, Vietnam.
- **First Prize**, Best Poster Award (Physics), St. Stephen's College (Feb 2025); **Second Prize** for the same poster at Kirori Mal College and Hindu College, University of Delhi. *Topic: Quantum Entanglement and Two-Flavor Neutrino Oscillations in Vacuum and Matter.*

EXTRACURRICULAR ACTIVITIES

- **Research Student, Center for Disability Research and Training (CDRT)** *Aug 2024 – Aug 2025*
  Led Physics team in a STEM outreach project for persons with disabilities; developed a syllabus for visually impaired students (SEC), designed survey instruments, and attended an IIT Delhi project-development workshop.
- **Member, KMC North East Cell Football Team** *Aug 2023 – Aug 2024*
  Represented Kirori Mal College in inter-college football tournaments in my 1st year, building teamwork, discipline, and time management alongside academics.